

Conformalized-KANs: Uncertainty Quantification with Coverage Guarantees for Kolmogorov-Arnold Networks (KANs) in Scientific Machine Learning

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Abstract

This paper explores uncertainty quantification (UQ) methods in the context of Kolmogorov–Arnold Networks (KANs). We apply an ensemble approach to KANs to obtain a heuristic measure of UQ, enhancing interpretability and robustness in modeling complex functions. Building on this, we introduce Conformalized-KANs, which integrate conformal prediction, a distribution-free UQ technique, with KAN ensembles to generate calibrated prediction intervals with guaranteed coverage. Extensive numerical experiments are conducted to evaluate the effectiveness of these methods, focusing particularly on the robustness and accuracy of the prediction intervals under various hyperparameter settings. We show that the conformal KAN predictions can be applied to recent extensions of KANs, including Finite Basis KANs (FBKANs) and multifidelity KANs (MFKANs). The results demonstrate the potential of our approaches to significantly improve the reliability and applicability of KANs in scientific machine learning.

¹ [1Equal contribution.]

1 Introduction

Scientific machine learning (SciML) merges machine learning with scientific computing to address complex problems in fields such as physics, fluids, and biology [1, 2]. While promising, the efficacy of SciML is often hampered by the extensive data requirements of traditional neural networks, limiting their use in data-sparse scientific domains. Thus, there is a critical need for innovative neural network architectures that can operate effectively with limited data and adhere to the scientific principles of the datasets, enhancing model efficiency and applicability across various scientific disciplines.

Kolmogorov-Arnold Networks (KANs) [3, 4] are a class of neural architectures inspired by the Kolmogorov-Arnold representation theorem [5], which posits that any multivariate continuous function can be represented as a superposition of continuous functions of one variable. KANs leverage this theoretical foundation to model complex, high-dimensional functions efficiently. By structuring the network to decompose high-dimensional inputs into simpler, univariate functions before recombining them, KANs can offer advantages over multilayer perceptrons (MLPs), such as discovering interpretable models. However, KANs can sometimes struggle to achieve the accuracy of MLPs [6, 7]. Numerous variants of KANs have appeared in the literature, including physics-informed KANs (PIKANs) [8, 9, 10, 11], temporal KANs [12], wavelet KANs [13, 14, 15], graph KANs [16, 17, 18], Chebyshev KANs (cKANs) [19, 20], fractional KANs [21], and deep operator KANs [22]. KANs have been applied to time-series analysis [23], fluid dynamics [24, 25], and computer vision [26, 27], among many other applications.

In our previous work, we developed Finite Basis KANs (FBKANs) [28] and Multi-Fidelity KANs (MFKANs) [29]. FBKANs are a domain decomposition-based network architecture that enables the parallel training of several small KANs, which are only coupled within overlapping regions, providing accurate solutions for multiscale problems. For MFKANs, we implemented a training strategy that leverages a pre-trained low-fidelity model in conjunction with a small amount of high-fidelity data to enhance high-fidelity predictions. Both FBKANs and MFKANs have demonstrated accuracy and robustness when sufficient training data is available. As shown for physics-informed neural networks (PINNs) [30] in [31], both approaches may also be combined for further improvements.

Insufficient training data often leads to inaccurate point predictions in SciML, underlining the critical need for effective uncertainty quantification (UQ) [32]. Traditional UQ approaches for SciML have primarily relied on ensemble [33] and Bayesian methods [34, 35, 36, 37, 38, 39], which provide heuristic estimations of uncertainty. However, these methods do not always offer definitive guarantees on the reliability of the predictions. In response to this limitation, our previous work has advanced the application of conformal prediction [40, 41, 42] to SciML, particularly in the realm of deep operator learning [43, 44]. Conformal prediction is a distribution-free UQ method that ensures coverage guarantees, thereby delivering robust prediction intervals. This approach significantly enhances prediction reliability in data-scarce scenarios, meeting the high standards required for credible and provable UQ in scientific applications. Building on this foundation, our objective in this paper is to combine the conformal prediction technique with KANs and their variants, including FBKANs and MFKANs, to enable uncertainty-aware predictions.

The main contributions of our work are:

- We integrate an ensemble approach with KANs to obtain a heuristic measure for UQ, which is compatible with KAN variants such as FBKANs and MFKANs.
- Leveraging this UQ measure, we combine conformal prediction, a supervised and distribution-free UQ method, with KAN ensembles to generate calibrated prediction intervals with user-defined coverage guarantees.
- We provide extensive experiments that test the properties of these prediction intervals under various hyperparameter configurations.

The remainder of the paper is organized as follows. Section 2 provides background information on KANs, FBKANs, and MFKANs. Section 3.1 develops an ensemble method for obtaining a heuristic measure of uncertainty. We then describe the conformal prediction UQ method tailored for KANs, FBKANs, and MFKANs, designed to deliver reliable prediction intervals with coverage guarantees in Section 3.2.

Numerical experiments demonstrating the properties and efficacy of the conformal prediction method are presented in Section 4. Finally, Section 5 concludes the paper.

2 Background

This section provides an overview of KANs, FBKANs, and MFKANs. For a detailed description, we refer the reader to [28, 29].

2.1 Kolmogorov-Arnold Networks (KANs)

KANs [3, 4] leverage the Kolmogorov-Arnold representation theorem [5] to efficiently approximate multivariate functions using neural network architectures. According to this theorem, any continuous multivariate function can be represented as a finite sum of compositions involving continuous univariate functions. KANs translate this theorem into a flexible architecture designed to learn complex nonlinear mappings effectively.

Formally, given an input vector $\mathbf{x} \in \mathbb{R}^{m_0}$, a target function $f(\mathbf{x})$ is approximated by the nested composition:

$$f(\mathbf{x}) \approx \sum_{i_{n_l-1}=1}^{m_{n_l-1}} \varphi_{n_l-1, i_{n_l-1}} \left(\sum_{i_{n_l-2}=1}^{m_{n_l-2}} \cdots \left(\sum_{i_2=1}^{m_2} \varphi_{2, i_2} \left(\sum_{i_1=1}^{m_1} \varphi_{1, i_1} \left(\sum_{i_0=1}^{m_0} \varphi_{0, i_0} (x_{i_0}) \right) \right) \right) \right) \right), \quad (1)$$

$$\triangleq \mathcal{K}(\mathbf{x}).$$

In this formulation, n_l denotes the number of layers, $\{m_j\}_{j=0}^{n_l}$ represents the number of neurons in each layer, and $\varphi_{i,j,k}$ are univariate activation functions. These activation functions are defined as polynomial functions of degree k , represented on a grid with g points, and consist of a weighted combination of a basis function $b(x)$ and a B-spline:

$$\varphi(x) = w_b b(x) + w_s \text{spline}(x),$$

where

$$b(x) = \frac{x}{1 + e^{-x}} \quad \text{and} \quad \text{spline}(x) = \sum_i c_i B_i(x).$$

Here, $B_i(x)$ are polynomial B-splines of degree k , and c_i , w_b , and w_s represent learnable parameters.

KANs evaluate these B-splines on a pre-defined grid. Specifically, for a one-dimensional domain $[a, b]$, the grid is partitioned into g_1 intervals with grid points $\{t_0 = a, t_1, t_2, \dots, t_{g_1} = b\}$, following the approach detailed in [3]. Moreover, the grid extension method from [3] enables transferring coarse-grained spline representations onto a finer grid, enhancing the network's expressive capability and flexibility.

Next, we apply domain decomposition to obtain FBKANs [28]. We focus on the mean squared error in the loss function, rather than the sparsity-enforcing method described in [3]. While recent work has proposed many variations of KANs, we adopt the formulation outlined in [3] for this study. However, the techniques developed in this paper, including the domain decomposition method we present next, can be applied to many KAN variants.

2.2 Finite Basis KANs (FBKANs)

Finite basis KANs (FBKANs) [28] are inspired by the domain decomposition approach in finite basis physics-informed neural networks (FBPINNs) developed in [45] and extended in [46, 47, 48]. In particular, FBPINNs provide a domain decomposition-based architecture for physics-informed neural networks (PINNs) [30]. In FBKANs, we employ partition of unity functions $\{\omega_j\}_{j=1,\dots,L}$ based on an overlapping domain decomposition, such that the function approximation in equation (1) takes the following form:

$$\begin{aligned} f(\mathbf{x}) &\approx \sum_{j=1}^L w_j(\mathbf{x}) \left[\sum_{i_{n_l-1}=1}^{m_{n_l-1}} \varphi_{n_l-1, i_{n_l}, i_{n_l-1}}^j \left(\sum_{i_{n_l-2}=1}^{m_{n_l-2}} \dots \left(\sum_{i_2=1}^{m_2} \varphi_{2, i_3, i_2}^j \left(\sum_{i_1=1}^{m_1} \varphi_{1, i_2, i_1}^j \left(\sum_{i_0=1}^{m_0} \varphi_{0, i_1, i_0}^j(x_{i_0}) \right) \right) \right) \dots \right) \right] \\ &\triangleq \sum_{j=1}^L w_j(\mathbf{x}) \mathcal{K}^j(\mathbf{x}, \theta^j). \end{aligned} \quad (2)$$

Here, $\mathcal{K}^j(x, \theta_j)$ represents the j -th KAN with trainable parameters θ_j .

The partition of unity functions is obtained as follows. We decompose the domain Ω into overlapping subdomains. Each subdomain Ω_j corresponds to the interior of the support of the function ω_j , and the set of functions ω_j forms a partition of unity. Specifically, for L overlapping subdomains, we have

$$\Omega = \bigcup_{j=1}^L \Omega_j, \quad \text{supp}(\omega_j) = \bar{\Omega}_j, \quad \text{and} \quad \sum_{j=1}^L \omega_j = 1 \quad \text{in } \Omega.$$

We can define the partition of unity functions in several ways. In this work, we construct them based on the expression:

$$w_j = \frac{\hat{w}_j}{\sum_{j=1}^L \hat{w}_j},$$

where, in one dimension,

$$\hat{w}_j = \begin{cases} 1, & L = 1, \\ [1 + \cos(\pi(x_i - \mu_{ij}) / \sigma_{ij})]^2, & L > 1. \end{cases}$$

Here, μ_{ij} , and σ_{ij} represent the center and half-width of each subdomain j in each direction and $x \in \mathbb{R}^d$. As has been investigated for FBPINNs in [46], the width of the overlap between subdomains has an impact on the model performance; generally, the performance is improved if the width of the overlap is increased.

To train the trainable parameters $\theta = \{\theta^j\}_{j=1}^L$ of FBKANs, we use the dataset of tuples $\mathcal{D} = \{\mathbf{x}^i, f(\mathbf{x}^i)\}_{i=1}^N$ to minimize the following data-driven loss function:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \left(\sum_{j=1}^L w_j(\mathbf{x}^i) \mathcal{K}^j(\mathbf{x}^i, \theta^j) - f(\mathbf{x}^i) \right)^2.$$

This means that we simply insert the FBKAN architecture in (2) into the mean squared error loss function.

2.3 Multi-Fidelity KANs (MFKANs)

We follow [29] and draw inspiration from composite multifidelity neural networks [49], which consist of three networks. The first network learns from the low-fidelity training data. The second and third networks combine to capture the correlation between the low-fidelity model’s output and the high-fidelity data, providing an accurate high-fidelity prediction. Specifically, the second network models the linear correlation, while the third network captures the nonlinear correlation. Ideally, the low- and high-fidelity datasets are highly correlated, so the linear correlation explains most of the relationship, with the nonlinear correlation acting as a small adjustment. This structure makes the nonlinear network smaller and improves training robustness, especially when limited high-fidelity data is available.

Thus, we model MFKANs using three components: a low-fidelity KAN ($\mathcal{K}_L$), a linear KAN ($\mathcal{K}_l$), and a nonlinear KAN ($\mathcal{K}_{nl}$). The low-fidelity KAN approximates the low-fidelity data. Its output, combined with the original input, serves as input to both the linear and nonlinear KANs. The linear KAN, acting as a simple linear mapping, captures linear correlations among these combined inputs and the high-fidelity data. It is implemented without any hidden layers and uses first-degree polynomial basis functions, making it suitable for learning only global linear trends. In contrast, the nonlinear KAN is specifically designed to model and capture residual nonlinearities that the linear KAN inherently cannot represent, thus refining the prediction further. This is enabled by using higher-degree polynomial basis functions and additional hidden layers in the nonlinear KAN, which increase its expressive power beyond that of the linear KAN.

Let us consider two datasets: a low-fidelity dataset with labeled pairs $\{(\mathbf{x}_i, f_L(\mathbf{x}_i))\}_{i=1}^{N_{LF}}$ and a high-fidelity dataset $\{(\mathbf{x}_j, f_H(\mathbf{x}_j))\}_{j=1}^{N_{HF}}$. Following [29], we note that $\{\mathbf{x}_j\}_{j=1}^{N_{HF}}$ does not need to be a subset of $\{\mathbf{x}_i\}_{i=1}^{N_{LF}}$.

Low-Fidelity KAN. We pretrain a low-fidelity KAN and freeze its weights during high-fidelity KAN fine-tuning. To train the parameters θ_L of the low-fidelity KAN, we optimize the following loss function using the low-fidelity dataset:

$$\mathcal{L}(\theta_L) = \frac{1}{N_{LF}} \sum_{i=1}^{N_{LF}} (\mathcal{K}_L(\mathbf{x}_i, \theta_L) - f_L(\mathbf{x}_i))^2.$$

High-Fidelity KAN. Following [29], the high-fidelity KAN prediction is a convex combination of the linear and nonlinear KANs expressed as:

$$\mathcal{K}_H(\mathbf{x}) = \alpha \mathcal{K}_{nl}(\mathbf{x}) + (1 - \alpha) \mathcal{K}_l(\mathbf{x}).$$

Here, α is a trainable parameter. We train the high-fidelity KAN parameters $\theta_H = \{\theta_l, \theta_{nl}\}$ using the high-fidelity dataset to minimize the following loss function:

$$\mathcal{L}_H(\theta_H, \alpha) = \sum_{j=1}^{N_{HF}} (\mathcal{K}_H(\mathbf{x}_j, \theta_H) - f_H(\mathbf{x}_j))^2 + \lambda_\alpha \alpha^n + w \sum_{l=0}^{L-1} \|\Phi_{nl}\|. \quad (3)$$

In the above, n is a scalar hyper-parameter, typically set to $n = 4$. This value minimizes α , encouraging the method to focus on learning the maximum linear correlation. However, if n is too small, α becomes too small, preventing the method from capturing the full nonlinear correlation, which limits its expressiveness. If a strong nonlinear correlation exists between the high- and low-fidelity data, a larger value of n may be chosen, though this may require more high-fidelity data for training. Similarly, the hyper-parameter λ_α is also selected before training. The final term in the above loss function represents the sum of the mean squared values of each KAN layer in the nonlinear network, inspired by [29], and is given by

$$\|\Phi_{nl}\| = \frac{1}{n_{in}n_{out}} \sum_{i=1}^{n_{in}} \sum_{j=1}^{n_{out}} |\phi_{i,j}^{nl}|^2.$$

Finally, we set $w = 0$ or $w = 1$ based on any known correlation between the low-fidelity and high-fidelity data. If we expect the correlation to be more strongly linear, a larger value of w forces learning a larger linear correlation.

3 Uncertainty Quantification Methodology

KANs, including FBKANs and MFKANs, provide only point predictions, which may be inaccurate. Since SciML applications demand robust solutions, we must rigorously estimate model uncertainty. In this section, we first introduce ensemble KANs to empirically measure uncertainty. Then, we apply split conformal prediction—a supervised, distribution-free UQ method—to generate rigorous prediction intervals for KANs using the ensemble predictions.

3.1 Ensemble of KANs

This section develops an ensemble of *KAN-based networks* to enhance prediction accuracy and model robustness. To achieve this, we train an ensemble of models, each initialized randomly, to capture model uncertainty and improve generalization.

Each model in the ensemble KANs is trained by optimizing the loss function, starting from a different random initialization denoted by θ_0 , where θ_0 represents the initial parameter state before training begins. After training an ensemble of M models, from different random initial conditions θ_0^j , we obtain the parameter set $\{\theta^j\}_{j=1}^M$. For a given test point x_{test} , we compute ensemble statistics by performing a forward pass through all M models. Specifically, we calculate the mean and standard deviation of the M -ensemble of KANs as follows:

$$\begin{aligned} \mu_M(x_{\text{test}}) &= \frac{1}{M} \sum_{j=1}^M \mathcal{K}(x_{\text{test}}, \theta^j), \\ \sigma_M(x_{\text{test}}) &= \sqrt{\frac{1}{M} \sum_{j=1}^M \left(\mathcal{K}(x_{\text{test}}, \theta^j) - \mu_M(x_{\text{test}}) \right)^2}. \end{aligned}$$

Here, $\mathcal{K}(\cdot, \theta^j)$ represents the forward pass of a single model in the M -ensemble, $\mu_M(\cdot)$ denotes the mean prediction of the ensemble, and $\sigma_M(\cdot)$ corresponds to the standard deviation.

Constructing an ensemble of FBKANs and MFKANs follows a similar procedure, but in the multi-fidelity case, we begin by freezing a single low-fidelity KAN and then train the ensemble MFKANs by optimizing the high-fidelity loss function (3), starting from a random initialization $\theta_H(0)$ while incorporating the frozen pretrained low-fidelity parameters θ_L .

To estimate uncertainty, we construct a 1.96-sigma prediction interval [33, 34] for a given test input x_{test} . First, we compute the mean $\mu_M(x_{\text{test}})$ and standard deviation $\sigma_M(x_{\text{test}})$ of the ensemble predictions from the trained M -ensemble models. The 1.96-sigma prediction interval is then defined as the interval:

$$[\mu_M(x_{\text{test}}) - 1.96\sigma_M(x_{\text{test}}), \mu_M(x_{\text{test}}) + 1.96\sigma_M(x_{\text{test}})].$$

This set captures the range within which the true output $f(x_{\text{test}})$ is expected to lie with approximately 95% confidence, assuming the predictions follow a normal distribution.

In SciML, assuming that model predictions follow a normal distribution is often incorrect. Consequently, providing a 95% coverage guarantee for ensemble predictions is inherently difficult. The uncertainty in real-world datasets, especially in those corresponding to complex scientific problems, often deviates from the assumptions of normality. To address this challenge, we turn to split conformal prediction in the next section. This method leverages the ensemble predictions to generate prediction intervals with rigorous coverage guarantees, ensuring more reliable uncertainty estimation without relying on normality assumptions.

3.2 Conformal Prediction for KANs

Conformal prediction [40, 41, 42] offers a robust framework for creating prediction intervals with guaranteed coverage, providing a distribution-free approach to UQ in machine learning models. This method is particularly valuable for SciML applications where data distributions often defy conventional assumptions. At its core, conformal prediction ensures that a prediction interval includes the true output with a predefined probability, independent of the underlying data distribution. This attribute establishes conformal prediction as a versatile and reliable tool in scenarios where traditional statistical methods fall short.

To build these prediction intervals, conformal prediction leverages a model trained on a specific dataset to generate a heuristic measure of uncertainty. For our purposes, we use the standard deviation of the ensemble’s predictions as this measure of uncertainty.

Algorithm 1 outlines the procedure for applying conformal prediction to ensemble KANs, including ensemble FBKANs and MFKANs. In particular, to apply conformal prediction effectively we require a *calibration dataset* that must be exchangeable. This exchangeability ensures that the order of data points does not impact the prediction interval, thereby ensuring the model’s conformal predictions remain consistent and reliable for new, unseen data. Notably, since i.i.d. data inherently implies exchangeability, assuming i.i.d. data in the calibration set guarantees that the predictions are valid across the dataset. In our configuration, the calibration dataset comprises n data pairs, $\mathcal{D}_{\text{cal}} = \{x_j, f(x_j)\}_{j=1}^n$, where x_j is the input and $f(x_j)$ is the corresponding output. These tuples are essential for estimating the non-conformity scores, which are crucial for generating accurate prediction intervals.

Algorithm 1 Split Conformal Prediction for KAN-based model

- 1: **Input:** Calibration data $\mathcal{D}_{\text{cal}} = \{x_j, f(x_j)\}_{j=1}^n$, miscoverage level α , and trained ensemble KAN-based model μ_M, σ_M .
- 2: **Output:** Confidence intervals for new test examples.
- 3: Use ensemble KAN-based model to predict the mean and standard deviations $\mu_M(x_j), \sigma_M(x_j)$.
- 4: Calculate nonconformity scores $s_1, s_2, \dots, s_n$ for $\mathcal{D}_{\text{cal}}$, where:

$$s_j = \frac{|f(x_j) - \mu_M(x_j)|}{\sigma(x_j)}.$$

- 5: Determine the empirical $[(n+1)(1-\alpha)] / n$ -quantile of the set of scores, denote as $\hat{q}_\alpha$.
- 6: **for** each new test point x_{test} **do**
- 7: Obtain from ensemble KANs-based models $\mu_M(x_{\text{test}}), \sigma_M(x_{\text{test}})$.
- 8: Construct the prediction interval:

$$\mathcal{C}_\alpha(x_{\text{test}}) = [\mu_M(x_{\text{test}}) - \hat{q}_\alpha \cdot \sigma_M(x_{\text{test}}), \mu_M(x_{\text{test}}) + \hat{q}_\alpha \cdot \sigma_M(x_{\text{test}})].$$

- 9: **end for**
- 10: **return** Confidence intervals $\mathcal{C}_\alpha(x_{\text{test}})$ ensuring at least $1 - \alpha$ coverage.

We use the calibration dataset $\mathcal{D}_{\text{cal}}$ to compute nonconformity scores $s_1, \dots, s_n$. These scores evaluate the degree to which each data point in the calibration set aligns with the predictions of the trained KAN-based model, considering the inherent uncertainty. Essentially, a data point with a low nonconformity score indicates good alignment between the KAN's predictions and the actual outcomes, suggesting its inclusion in the prediction interval. In ensemble-based models like ours, we estimate this uncertainty by calculating the standard deviation of the ensemble's predictions. Therefore, for each tuple $(x_j, f(x_j))$ in $\mathcal{D}_{\text{cal}}$, we determine the j -th nonconformity score s_j as follows [41, 43, 44]:

$$s_j = \frac{|f(x_j) - \mu_M(x_j)|}{\sigma(x_j)}.$$

To obtain the prediction interval, we compute $\hat{q}_\alpha$ as the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile of the set of nonconformity scores $s_1, \dots, s_n$. Using this quantile $\hat{q}_\alpha$, for a new test input x_{test} , the proposed conformal prediction method for KANs produces a prediction interval:

$$\mathcal{C}_\alpha(x_{\text{test}}) = [\mu_M(x_{\text{test}}) - \hat{q}_\alpha \cdot \sigma_M(x_{\text{test}}), \mu_M(x_{\text{test}}) + \hat{q}_\alpha \cdot \sigma(x_{\text{test}})].$$

Here, the prediction interval is ideally small and includes the target $f(x_{\text{test}})$ with a high probability of $1 - \alpha$, as selected by the user.

The theoretical guarantees for the prediction intervals generated by the proposed conformal prediction method for KANs are outlined next. Given that the data is exchangeable, the method assures coverage:

$$\mathbb{P}(f(x_{\text{test}}) \in \mathcal{C}_\alpha(x_{\text{test}})) \geq 1 - \alpha,$$

Note that while i.i.d. random variables are necessarily exchangeable, the converse may not hold. Therefore, our datasets in the experiments are always i.i.d.. For prediction intervals of the form $\mathcal{C}_\alpha = \{y = f(x_{\text{test}}) : s(x_{\text{test}}, y) \leq \hat{q}_\alpha\}$, including as proposed in this paper, the theorem in [40, 42] ensures coverage guarantees of the form:

$$1 - \alpha \leq \mathbb{P}(f(x_{\text{test}}) \in \mathcal{C}_\alpha(x_{\text{test}})) \leq 1 - \alpha + \frac{1}{n+1}.$$

Figure 1 illustrates the schematic of the Conformalized-KANs framework. Within this framework, the ensemble KANs can comprise any ensemble of trained KAN models, including FBKAN and MFKAN.

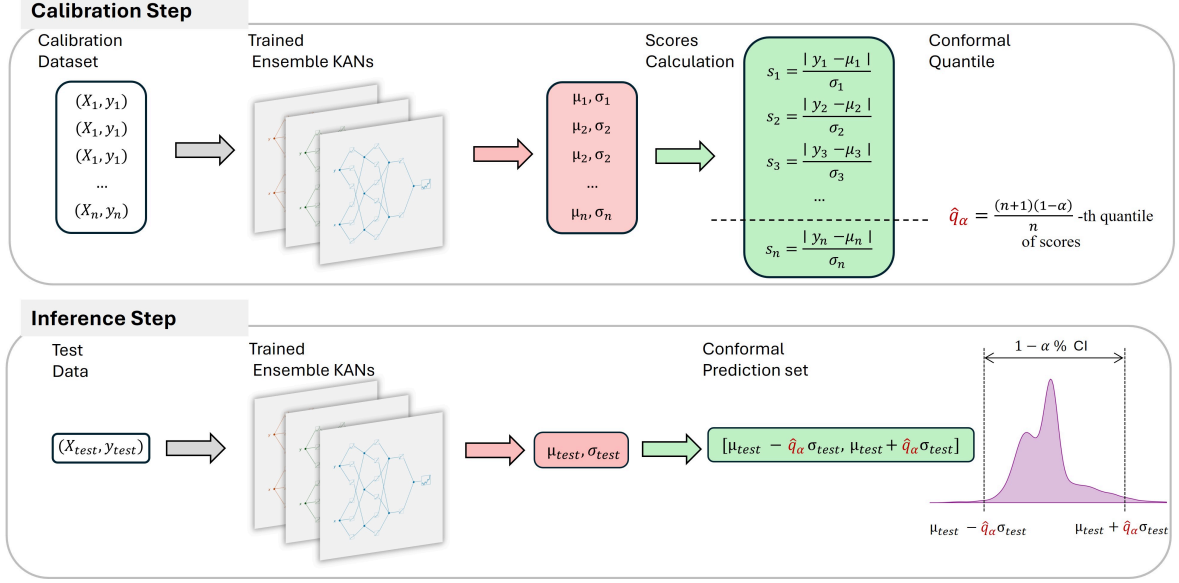


Figure 1: Schematic representation of the Conformalized-KANs framework. **Calibration step:** A dedicated calibration dataset is used to estimate the conformal quantile $\hat{q}_\alpha$ corresponding to a target miscoverage rate α . **Inference step:** This quantile is then employed to construct prediction intervals that achieve the desired coverage.

4 Numerical Experiments

This section evaluates the performance of the proposed Conformalized-KANs in ensuring guaranteed coverage. The effectiveness of this approach is demonstrated through four approximation tasks: a 1-D function, a 2-D function, a multi-fidelity problem, and a PDE problem. In each case, an ensemble of KAN networks is trained to generate a heuristic measure of uncertainty and provide prediction intervals. Conformal prediction is then applied to refine these intervals, ensuring the desired coverage level. Across all experiments, the objective is to predict the 95% confidence interval, corresponding to a miscoverage rate of $\alpha = 0.05$.

4.1 Metrics

To evaluate the performance of the proposed Conformalized-KANs, we use the following metrics.

Average Coverage: This metric quantifies the proportion of observed values that fall within the predicted intervals, reflecting the model's reliability. It is defined as:

$$\text{Average Coverage} = \frac{1}{N} \sum_{i=1}^N I(y_i \in [\hat{y}_{i, \text{lower}}, \hat{y}_{i, \text{upper}}]),$$

where N is the total number of data points, $y_i = f(x_i)$ is the observed value, and $[\hat{y}_{i, \text{lower}}, \hat{y}_{i, \text{upper}}]$ represents the predicted interval. The indicator function $I(\cdot)$ returns 1 if y_i falls within the interval and 0 otherwise. A higher Average Coverage indicates better reliability in uncertainty estimation.

Prediction Interval Width (PIW): Average PIW represents the average width of predicted intervals, providing insight into model uncertainty:

$$\text{Average PIW} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_{i,\text{upper}} - \hat{y}_{i,\text{lower}}|.$$

A narrower Average PIW suggests more confident predictions, but overly tight intervals may lead to insufficient coverage. Ideally, Average PIW should be minimized while maintaining high Average Coverage to balance accuracy and UQ.

4.2 Experiment 1: 1-D Function

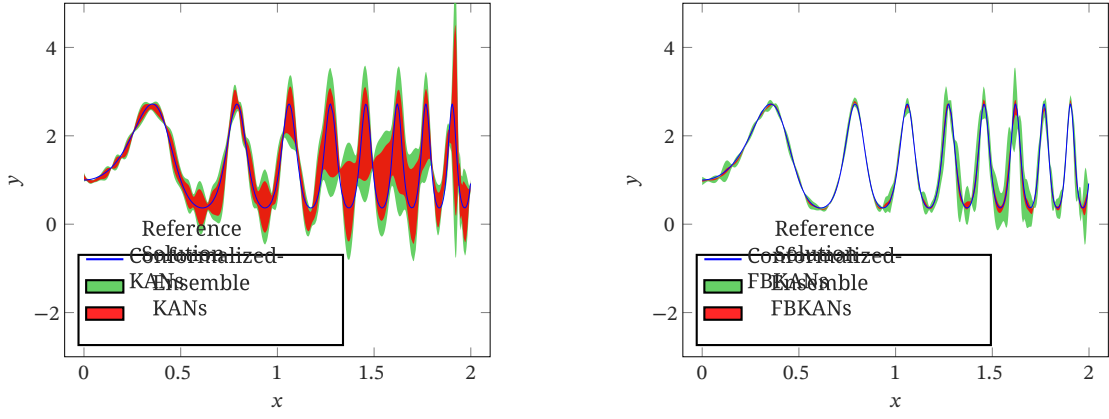
In this experiment, we consider the following 1-D function:

$$f(x) = \exp(\sin(0.3\pi x^2)),$$

on $x \in [0, 2]$. This function was selected due to its nonlinear and locally varying structure, which presents a meaningful challenge for predictive modeling. The sinusoidal component introduces periodic variations, creating regions of differing smoothness, while the exponential transformation amplifies these variations, making the function more dynamic and highlighting areas of higher uncertainty. We test this case using both KAN and FBKAN networks.

KAN Design and Datasets. To generate prediction intervals, we use an ensemble of size 4 for both KAN and FBKAN networks. For FBKANs, the number of subdomains is set to 10. The dataset comprises 400 random samples for training, 500 for calibration in conformal prediction, and 1,000 for evaluation.

Results. Figure 2 shows the predicted intervals from the ensemble KANs and ensemble FBKANs, along with their conformalized prediction intervals for the test data. Table 1 summarizes the results for each method. Both Conformalized-KANs and Conformalized-FBKANs achieve the target 95% coverage, while the original, non-conformal models fail to meet the desired confidence level. Consequently, the ensemble method yields smaller average width. Moreover, the Conformalized-KANs intervals are notably wider, reflecting greater inherent uncertainty in approximations made by standard KAN architectures. Meanwhile, Conformalized-FBKANs intervals are narrower but sufficiently accurate due to FBKAN’s use of domain decomposition, enabling localized and more precise approximations (see also the ablation study in Section A.1).



(a) Ensemble KANs and Conformalized-KANs Prediction Intervals

(b) Ensemble FBKANs and Conformalized-FBKANs Prediction Intervals

Figure 2: Conformalized prediction intervals generated on 1-D function test dataset, computed with an ensemble size of 4 and a target miscoverage rate of $\alpha = 0.05$.

Model	Average Coverage	Average PIW	Std. Dev PIW
Conformalized-KANs	96.10%	1.06	0.91
Conformalized-FBKANs	95.30%	0.41	0.42
Ensemble KANs	88.30%	0.73	0.62
Ensemble FBKANs	52.81%	0.08	0.08

Table 1: Average coverage, average interval width, and the standard deviation of interval widths for ensemble KANs, ensemble FBKANs, and their conformalized counterparts, evaluated on 1-D function test dataset. Each method targets a 95% prediction interval, corresponding to a miscoverage rate of $\alpha = 0.05$.

4.3 Experiment 2: 2-D Function

In this experiment, we consider the following 2-D function:

$$f(x, y) = \sin(6\pi x^2) \sin(8\pi y^2),$$

on $(x, y) \in [0, 1] \times [0, 1]$.

KAN Design and Datasets. For this experiment, an ensemble of size 4 is used for KAN and FBKAN models, with the number of subdomains for FBKAN set to 4. The dataset comprises 2,000 samples for training, 1,000 for calibration, and 1000 for evaluation.

Results. Figure 3 presents the conformalized prediction intervals for both models, alongside the reference solution, which corresponds to the true values of the target function f evaluated over the domain. Table 2 summarizes the coverage and width results. As shown in the table, for the KAN network the ensemble model tends to underestimate uncertainty, while Conformalized-KANs achieves the tar-

get 95% coverage. For the FBKAN model, the ensemble version overestimates coverage, whereas its conformalized counterpart refines the predictions, bringing them close to the target coverage level. Consequently, in this case, the Conformalized-FBKANs produces narrower prediction intervals compared to its non-conformalized counterpart.

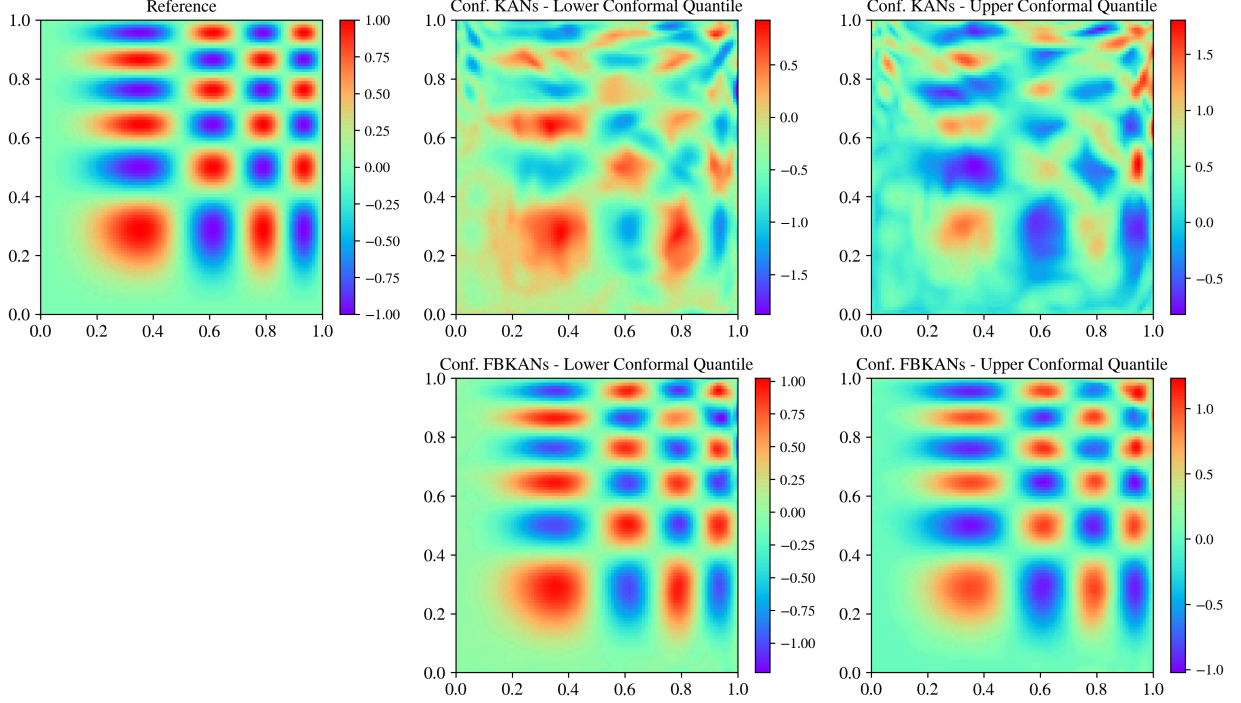


Figure 3: Conformalized prediction intervals generated by Conformalized-KANs and Conformalized-FBKANs on the 2-D function test dataset, computed with an ensemble size of 4 and a target miscoverage rate of $\alpha = 0.05$.

Model	mean Coverage	Average PIW	Std. Dev PIW
Conformalized-KANs	95.50%	0.62	0.37
Conformalized-FBKANs	94.85%	0.09	0.10
Ensemble KANs	92.44%	0.49	0.30
Ensemble FBKANs	97.07%	0.10	0.12

Table 2: Average coverage, average interval width, and the standard deviation of interval widths for ensemble KANs, ensemble FBKANs, and their conformalized counterparts, evaluated on 2-D function test dataset. Each method targets a 95% prediction interval, corresponding to a miscoverage rate of $\alpha = 0.05$.

4.4 Experiment 3: Multi-Fidelity Problem

In this experiment, we consider a multi-fidelity problem involving a jump function with a linear correlation between the low- and high-fidelity expressions. The low-fidelity function $f_L(x)$ and the high-fidelity function $f_H(x)$ are defined as follows:

$$f_L(x) = \begin{cases} 0.1(0.5(6x - 2)^2 \sin(12x - 4) + 10(x - 0.5) - 5) & x \leq 0.5, \\ 0.1(0.5(6x - 2)^2 \sin(12x - 4) + 10(x - 0.5) - 2) & x > 0.5, \end{cases}$$

$$f_H(x) = 2f_L(x) - 2x + 2,$$

for $x \in [0, 1]$.

KAN Design and Datasets. To generate prediction intervals, we train an ensemble of size 5 following the procedure detailed in Section 3.1. The dataset consists of 120 low-fidelity training samples and 5 high-fidelity samples for training. Additionally, 40 samples are used for calibration, and 200 data points are allocated for evaluating the MFKAN models.

Results. Figure 4 illustrates the prediction intervals produced by the ensemble MFKANs, along with its conformalized predictions for the test dataset. Table 3 summarizes the performance metrics for each approach. Consistent with previous experiments, the Conformalized-MFKANs achieves the desired 95% coverage, whereas the original, non-conformal model does not, indicating underestimation of uncertainty. Consequently, the intervals from the conformalized model are wider on average, reflecting a more reliable representation of uncertainty (see also the ablation study in Section A.2).

Model	mean Coverage	Average PIW	Std. Dev PIW
Conformalized-MFKANs	96.01%	0.18	0.11
Ensemble MFKANs	90.00%	0.15	0.09

Table 3: Average coverage, average interval width, and the standard deviation of interval widths for ensemble MFKANs and its conformalized counterpart, evaluated the high-fidelity test dataset. Each method targets a 95% prediction interval, corresponding to a miscoverage rate of $\alpha = 0.05$.

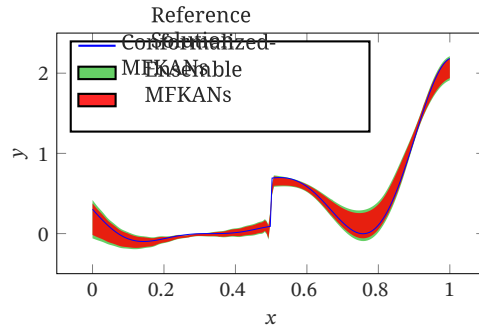


Figure 4: Conformalized prediction intervals generated by ensemble MFKANs and Conformalized-MFKANs on the high-fidelity test dataset, computed with an ensemble size of 5 and a target miscoverage rate of $\alpha = 0.05$.

4.5 Experiment 4: PDE problem

In this experiment, we analyze the wave equation, a second-order PDE, expressed as:

$$\frac{\partial^2 f}{\partial t^2} - c^2 \frac{\partial^2 f}{\partial x^2} = 0, \quad (x, t) \in [0, 1] \times [0, 1],$$

subject to the following boundary and initial conditions:

$$f(0, t) = 0, \quad f(1, t) = 0, \quad t \in [0, 1],$$

$$f(x, 0) = \sin(\pi x) + 0.5 \sin(4\pi x), \quad f_t(x, 0) = 0, \quad x \in [0, 1].$$

The exact solution is given by:

$$f(x, t) = \sin(\pi x) \cos(c\pi t) + 0.5 \sin(4\pi x) \cos(4c\pi t).$$

We examine the case where $c = \sqrt{2}$. To train the KAN-based models, we employ a physics-informed learning framework in which we enforce supervision through initial and boundary condition data, along with physical consistency via the PDE residual. The total loss function consists of four components: the initial condition loss $\mathcal{L}_{\text{IC}}$, which enforces consistency with the prescribed initial profile; the temporal derivative loss $\mathcal{L}_t$, which imposes the initial velocity condition $\frac{\partial f}{\partial t}(x, 0) = 0$; the boundary condition loss $\mathcal{L}_{\text{BC}}$; and the residual loss $\mathcal{L}_{\text{res}}$, which quantifies the violation of the governing PDE.

To prevent the residual term from dominating the optimization process and to ensure balanced contributions from all components, we apply a weighting factor $\lambda_{\text{res}} = 0.01$ to the residual loss. The resulting total loss function is defined as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{IC}} + \mathcal{L}_t + \mathcal{L}_{\text{BC}} + \lambda_{\text{res}} \cdot \mathcal{L}_{\text{res}}.$$

This formulation ensures that the model adheres to both empirical observations and the underlying physical laws, while maintaining numerical stability throughout training.

KAN Design and Datasets. In this experiment, an ensemble of size 10 is utilized for both KAN and FBKAN models, with the number of subdomains for FBKANs set to 4. The dataset includes 1000 samples for training, 1200 for calibration, and 10000 for evaluation.

Results. Figure 5 depicts the conformalized prediction intervals produced by the KAN and FBKAN models, along with the reference solution derived from the exact analytical expression of the wave equation across the spatiotemporal domain. Table 4 summarizes the corresponding coverage and width metrics. The results indicate that the ensemble models tend to underestimate uncertainty by producing narrower prediction intervals. In contrast, the conformalized versions correct this issue by widening the intervals appropriately, thereby achieving the target coverage of 95% (see also the ablation study in Section A.3).

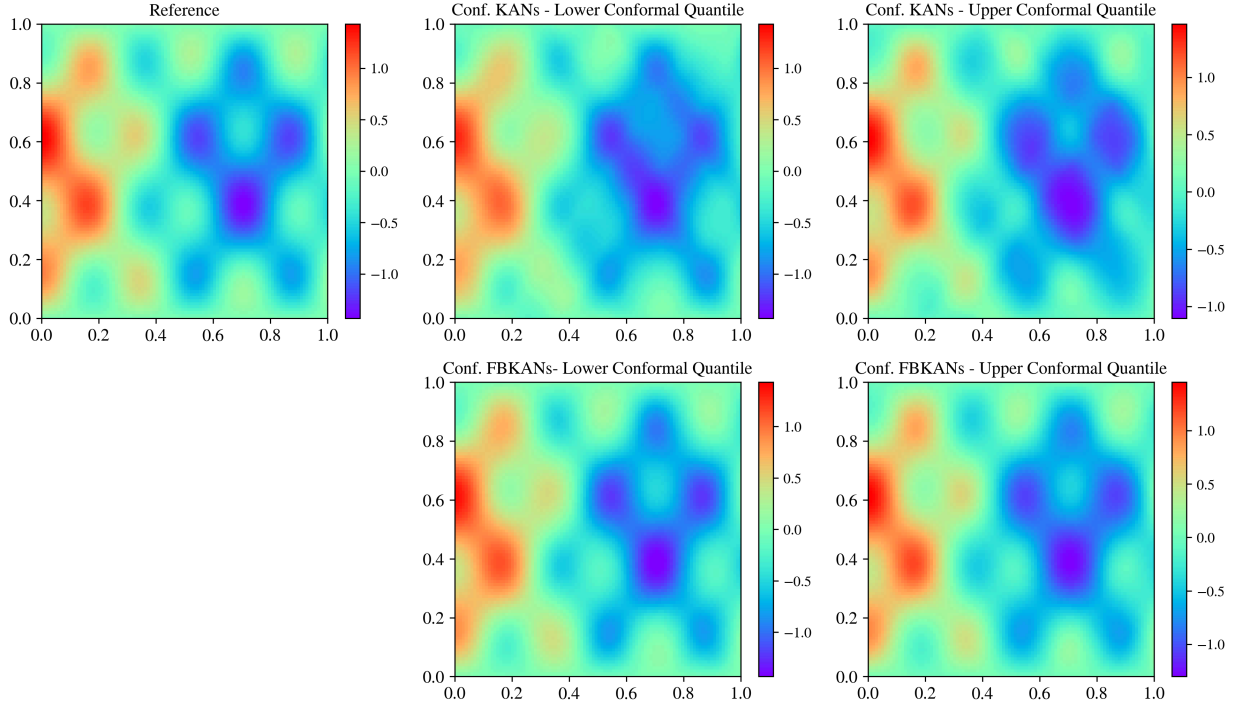


Figure 5: Conformalized prediction intervals generated by Conformalized-KANs and Conformalized-FBKANs on the wave equation test dataset, computed with an ensemble size of 10 and a target miscoverage rate of $\alpha = 0.05$.

Model	mean Coverage	Average PIW	Std. Dev PIW
Conformalized-KANs	95.44%	0.20	0.08
Conformalized-FBKANs	95.34%	0.06	0.03
Ensemble KANs	28.54%	0.06	0.02
Ensemble FBKANs	25.80%	0.01	0.007

Table 4: Average coverage, average width, and standard deviation of widths for ensemble KANs, ensemble FBKANs, and their conformalized versions, evaluated on the wave equation test dataset. Each method targets a 95% prediction interval, corresponding to a miscoverage rate of $\alpha = 0.05$

5 Conclusions

In this paper, we introduced Conformalized-KANs, a framework for UQ applied to KANs and their extensions, including FBKANs and MFKANs. By combining ensemble-based uncertainty estimation with conformal prediction, this approach generates prediction intervals with guaranteed coverage, without relying on restrictive distributional assumptions. Extensive numerical experiments across 1-D, 2-D, multi-fidelity, and PDE problems consistently demonstrated the effectiveness of Conformalized-KANs in guaranteeing the desired coverage of prediction intervals. Consequently, our results confirm that this framework significantly enhances the robustness and applicability of KAN models, addressing the crucial need for reliable UQ in data-limited scientific machine learning applications.

A notable outcome of our study is the consistent advantage of FBKANs over standard KANs in generating sharper prediction intervals. Even in their conformalized form, FBKANs produce significantly narrower intervals while still achieving the desired coverage guarantees, consistently outperforming KANs across all evaluated tasks. This underscores the strength of localized modeling via domain decomposition, which enables FBKANs to more accurately capture underlying function behavior and reduce epistemic uncertainty. Future work could build on these findings by incorporating adaptive subdomain partitioning strategies, allowing the model to dynamically adjust subdomain boundaries based on local uncertainty. This could potentially result in even sharper and better-calibrated prediction intervals. In addition, exploring advanced conformal prediction techniques, such as adaptive calibration sets, could enhance both the flexibility and efficiency of UQ in Conformalized-KANs.

Acknowledgments

This project was completed with support from the U.S. Department of Energy, Advanced Scientific Computing Research program, under the "Uncertainty Quantification for Multifidelity Operator Learning (MOLUcQ)" project (Project No. 81739). Pacific Northwest National Laboratory (PNNL) is a multi-program national laboratory operated for the U.S. Department of Energy (DOE) by Battelle Memorial Institute under Contract No. DE-AC05-76RL01830.

Appendix A Appendix: Ablation Studies

This appendix presents ablation studies for the experiments discussed in Section 4. We investigate the sensitivity of Conformalized-KANs and their variants (FBKANs and MFKANs) to key hyperparameters. These studies provide insight into the robustness and practical tuning of our approach across diverse learning tasks.

A.1 Experiment 1: 1-D Function

The Effect of Ensemble Size: We examine the impact of varying the ensemble size ($M \in \{3, 4, 5, 6, 7\}$) on the predicted intervals generated by ensemble KANs, ensemble FBKANs, and their conformalized counterparts. Figure 6(a) shows that both Conformalized-KANs and Conformalized-FBKANs consistently achieve coverage close to the target 95%, whereas their ensemble versions underestimate uncertainty.

Figure 6(b) illustrates that for Conformalized-KANs, increasing the ensemble size improves the accuracy of the conformalized networks, as reflected in the reduction of the mean and standard deviation of the PIW. Notably, with an ensemble size of seven, the width distributions for Conformalized-KANs outperform those of its ensemble counterpart. In contrast, as seen in Figure 6(c), the mean and standard deviation of PIW for FBKANs models remain stable as the ensemble size increases.

The Effect of the Number of Subdomains: We analyzed the effect of varying the number of subdomains ($L \in \{3, 6, 10, 15, 20\}$) on the predicted intervals for ensemble FBKANs and its conformalized counterpart. Figure 6(d) demonstrates that Conformalized-FBKANs consistently achieves the target 95% coverage, regardless of the number of subdomains. In contrast, the ensemble version consistently underestimates uncertainty across all tested configurations. This highlights the robustness of the Conformalized-FBKANs approach in providing reliable prediction intervals, even as problem complexity increases with a larger number of subdomains.

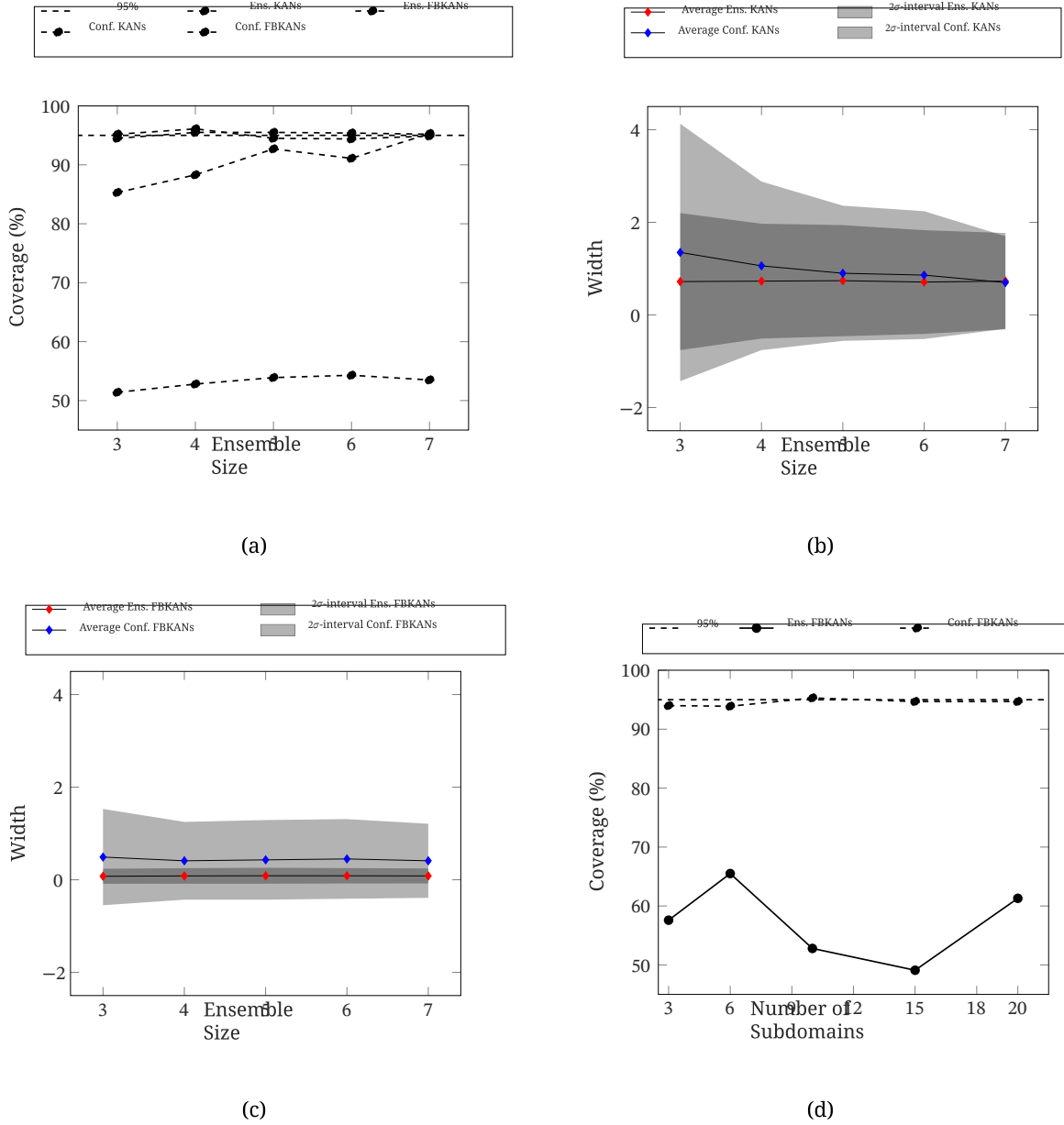


Figure 6: Experiment 1: 1-D Function. (a) Coverage performance of ensemble KANs and ensemble FBKANs, along with their conformalized counterparts, across different ensemble sizes ($M \in \{3, 4, 5, 6, 7\}$). (b) Width distributions for conformalized and ensemble KANs as a function of ensemble size. (c) Width distributions for conformalized and ensemble FBKANs as a function of ensemble size. (d) Coverage performance of ensemble FBKANs and its conformalized version across different subdomain counts ($L \in \{3, 6, 10, 15, 20\}$).

A.2 Experiment 3: Multi-Fidelity Problem

The Effect of the Ensemble Size: Figure 7(a) illustrates the effect of ensemble size ($M \in \{3, 4, 5, 6, 7\}$) on the prediction intervals of ensemble MFKAN and its conformalized counterpart. As shown, Conformalized-MFKANs maintains coverage close to the target 95%, whereas its ensemble version falls short. However, as depicted in Figure 7(b), achieving the desired coverage with conformalization comes at the cost of wider prediction intervals.

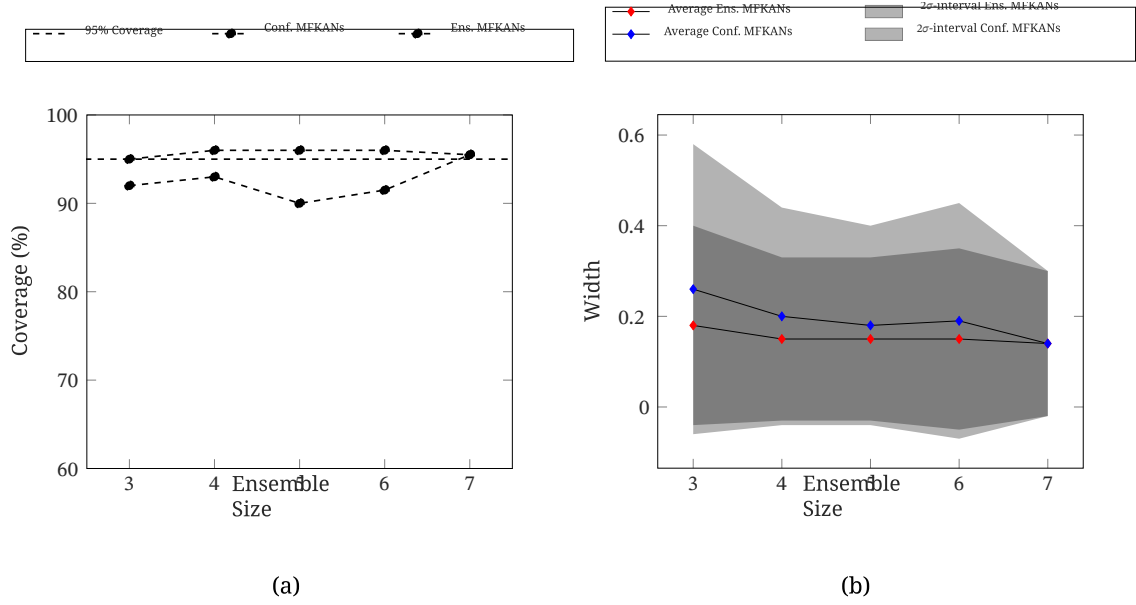


Figure 7: Experiment 3: Multi-Fidelity Problem. (a) Coverage performance of ensemble MFKANs, along with its conformalized version, across different ensemble sizes ($M \in \{3, 4, 5, 6, 7\}$). (b) Width distributions for ensemble and conformalized MFKANs as a function of ensemble size.

A.3 Experiment 4: PDE problem

The Effect of the Ensemble Size: Figure 8(a) illustrates the impact of varying the ensemble size ($M \in \{3, 4, 5, 6, 7, 8, 9, 10\}$) on the predicted intervals for ensemble KANs, ensemble FBKANs, and their conformalized counterparts. The results show that ensemble models yield lower coverage, resulting in underestimated uncertainty. In contrast, the conformalized models refine the predictions, maintaining coverage near the target 95% level.

The Effect of the Size of the Calibration Datasets: Figure 8(b) illustrates the impact of calibration dataset size on the coverage probability of conformalized-KANs and conformalized-FBKANs for an ensemble of 5. As shown in the figure, smaller calibration datasets lead to fluctuations in coverage, deviating from the target 95% level. However, as the dataset size increases, these variations diminish, resulting in more stable coverage that aligns with the desired confidence level. This highlights the crucial role of adequate calibration data in ensuring reliable prediction confidence.

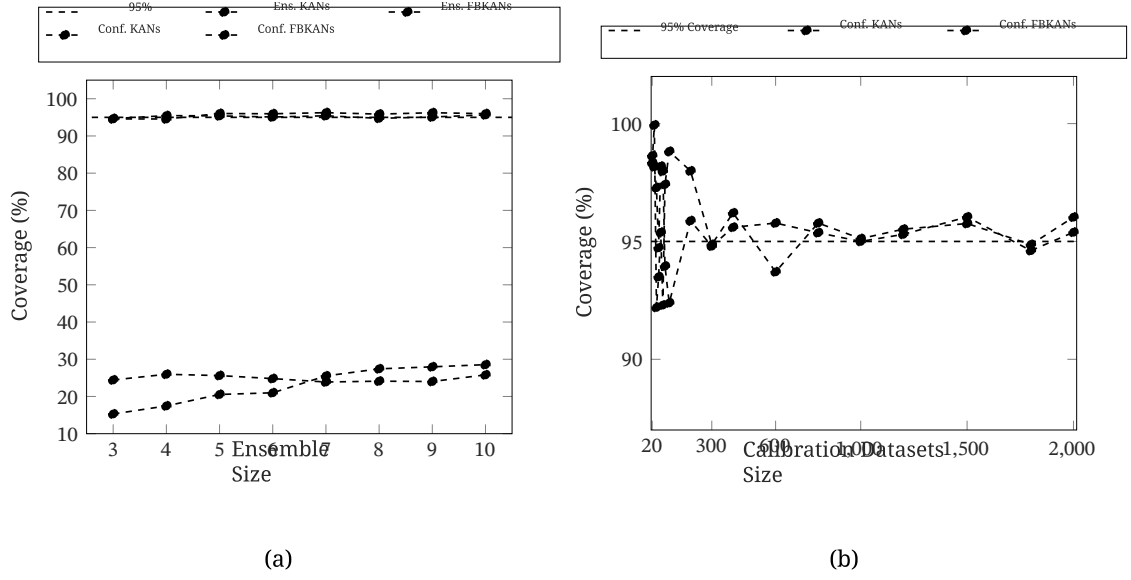


Figure 8: Experiment 4: PDE Problem. (a) Coverage performance of ensemble KANs and ensemble FBKANs, along with their conformalized counterparts, across different ensemble sizes ($M \in \{3, 4, 5, 6, 7, 8, 9, 10\}$). (b) Coverage performance of Conformalized-KANs and Conformalized-FBKANs with respect to calibration dataset size, evaluated using an ensemble of size 5.

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